

G. M. Mozahad

Location: Dhaka, Bangladesh * Phone: +8801886388416 * E-mail: moozaheed@gmail.com * Firm: craftsmenit.com

Website: moozaheed.github.io * GitHub: github.com/Moozaheed * LinkedIn: linkedin.com/in/moozaheed

Research Interests

Distributed & Federated Learning: Privacy-preserving ML, continual learning, non-IID optimization, decentralized and communication-efficient learning, differential privacy, secure aggregation.

Trustworthy & Explainable AI: Explainable AI (XAI), interpretable ML, robust and responsible AI, AI safety, adversarial robustness, mechanistic interpretability, privacy and security.

LLMs & Multi-Agent AI: Robust and reliable LLM pipelines, NLP, LLM security and evaluation, multi-agent systems, AI middleware architectures, Retrieval-Augmented Generation (RAG), context engineering.

Biomedical AI: Medical image analysis, clinical AI, domain adaptation, multimodal learning, computer-aided diagnosis (CAD), clinical foundation models, privacy-preserving healthcare.

Large-Scale Systems & Software Engineering: Distributed systems, cloud-native architectures, scalable microservices, fault tolerance, event-driven streaming pipelines, AI-assisted software engineering (AI-DLC).

Education

Bachelor of Science in Computer Science and Engineering

Department of Computer Science and Engineering

International Islamic University Chittagong

2020 – 2026

CGPA: 3.40 / 4.00

Medium of Instruction: English

Thesis: *Privacy-Preserving Federated Learning for Liver Disease Detection from Distributed Medical Records*

Honors & Awards: 3× Full Tuition Merit Waiver; Talent Development Program Award by WAMY.

Relevant Coursework: Machine Learning, Artificial Intelligence, Neural Networks, Computer Security, Digital Image Processing, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Numerical Methods, Mathematical Analysis, Discrete Mathematics, Statistics, Calculus.

Standardized Tests & Languages

IELTS Academic Target Overall Band: **6.5** (min. 6.0 each band)

Languages English (Full Professional & Academic Proficiency, CEFR C1), Bengali (Native)

Publications & Manuscripts Under Review

Privacy-Preserving Federated Continual Learning for Multi-Domain Medical Imaging Under Non-IID Skew

Manuscript Under Review

2026

- Formulated an end-to-end privacy-preserving federated continual learning framework to mitigate catastrophic forgetting and severe inter-site domain shifts in distributed clinical environments without raw patient data transfer.
- Evaluated systematically across multi-center medical imaging benchmarks under non-IID client partitions, achieving resilient convergence, robust diagnostic stability, and superior cross-domain generalization.

PIDM: A Middleware Framework for Detecting Prompt Injection Attacks in Multi-Agent LLM Pipelines

Manuscript Under Review

2026

- Architected a modular middleware security layer that intercepts, inspects, and neutralizes direct and indirect prompt injection attacks across inter-agent LLM communication pipelines.
- Developed real-time payload sanitization and semantic policy-checking mechanisms, safeguarding multi-agent collaboration and multi-step tool execution with minimal inference latency.

Academic & Teaching Experience

International Islamic University Chittagong (IIUC)

Undergraduate Teaching Assistant, Dept. of Computer Science & Engineering

June 2022 – January 2024

Chittagong, Bangladesh

- Mentored 150+ undergraduate students across core courses: **Data Structures, Algorithms, Object-Oriented Programming (OOP)**, and **Database Management Systems (DBMS)**.
- Conducted weekly laboratory sessions and problem-solving clinics, designed coding assessments, guided students through algorithmic analysis, and assisted faculty in evaluating semester capstone projects.

Professional Experience

Craftsmen IT

Founder & CEO | AI Systems, SaaS & Digital Transformation Consultancy

January 2026 – Present

Dhaka, Bangladesh

- Founded software consultancy delivering AI-native SaaS solutions, cloud architectures, and digital transformations.
- Architected advanced context engineering pipelines and multi-agent LLM workflows, reducing prompt latency by **40%**.
- Directed end-to-end technical architecture and product design for enterprise clients across the Gulf region.

Brain Station 23 PLC

Software Engineer I | Best Employee of the Quarter (Q1 2025)

April 2024 – Present

Dhaka, Bangladesh

- Forward-deployed engineer for enterprise client **Bloomex (Canada & Australia)**, architecting high-throughput GPMS, OMS, and ARCA APIs in Laravel Lumen to eliminate order data loss and cut query latency.
- Developed and maintained scalable SaaS applications using Node.js, PHP (Moodle), Electron.js, and Socket.io; applied modern **AI-DLC workflows** to accelerate testing and deployment cycles.
- Containerized microservices and automated CI/CD pipelines on AWS cloud infrastructure.

Selected Projects

Tiffin BD

Next.js, React Native, NestJS, TypeScript, MySQL, RabbitMQ, Socket.io, Docker, AWS, Terraform

- Pioneered an **AI-DLC workflow** leveraging multi-agentic AI to achieve **10x productivity gains**; engineered real-time asynchronous dispatch pipelines handling high-volume order queues via Socket.io, RabbitMQ, and Firebase.

Bloomex (Canada & Australia)

Laravel Lumen, React, Redux, Next.js, MySQL, Docker, AWS, Microservices

- Engineered high-throughput OMS, GPMS, and ARCA APIs across Canadian and Australian operations; optimized microservices to eliminate order data loss and reduce warehouse processing latency.

freemail.ai

Python, Postfix, Redis, Docker, High-Throughput REST APIs

- Architected disposable temporary email platform with dynamic multi-domain SMTP routing and auto-purge storage; engineered high-speed REST APIs handling real-time high-concurrency web traffic.

Moodle Proctoring Pro & Extra Restriction (AI Exam SaaS)

WebRTC, PHP, Node.js, Electron, MySQL, Docker, AWS, AI/ML

- Engineered open-source and SaaS exam proctoring platforms serving **2,000+ global institutions** and universities with official **Moodle Certified Integration**, WebRTC live streaming, face/object detection, and AI suspicion analysis.

Technical Skills

Programming: Python, C/C++, Java, JavaScript (ES6+), TypeScript, PHP, SQL, Bash, LaTeX

Machine Learning & AI: PyTorch, TensorFlow, Keras, Hugging Face, Scikit-learn, OpenCV, Deep Learning, Representation Learning, Transfer Learning, Multimodal Learning

Advanced Research: Federated & Continual Learning, Distributed Optimization, Context Engineering, Explainable AI (XAI), Trustworthy AI, Multi-Agent AI, LLMs, RAG, Prompt Injection Mitigation, AI Safety, Medical Image Analysis

Distributed & Cloud Systems: Microservices, REST APIs, WebSockets, WebRTC, Socket.IO, RabbitMQ, Redis, AWS Services, Fault-Tolerant Architecture, High-Concurrency Backends

Software Engineering: System Design, Scalable Architecture, API Design, Database Optimization, Design Patterns, Clean Architecture, Test Automation, CI/CD, Observability, Performance Engineering, AIDLC

DevOps & Cloud Infra: Docker, Kubernetes, Terraform (IaC), GitHub Actions, Git/GitHub, Linux (Ubuntu/Debian), CI/CD Pipelines, Cloud Deployment, Observability & Monitoring

Honors, Certifications & Activities

- **Contest Programming:** Participant, **ACM-ICPC Asia Regional**; solved **1,500+** problems on online judges.
- **Awards & Waivers:** **Best Employee of the Quarter**; **Full Tuition Fees Waiver**; **WAMY Scholarship**.
- **Certifications:** **Deep Learning with TensorFlow (IBM)**; **AI & Machine Learning with Python (BHTPA)**; **Evolution of AI Native Software Engineering (BS23)**; **Agentic Engineering (BS23)**.
- **Leadership & Service:** **Trainer & Organizing Secretary**, IIUC Competitive Programming Society; **Organizing Secretary**, IIUC Data Science Club; **Asst. General Secretary**, IIUC Computer Club.